

Exam Prep AI - Revision Plan

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Based on the provided information, I will create a study plan for you.

****High Priority Topics****

1. Recursion (module 3)
2. Dynamic programming and its advantages
3. Greedy algorithms and their characteristics
4. Brute-force and divide-and-conquer approaches
5. Algorithmic thinking and its practical applications
6. Pseudocode and flowchart representation of algorithms
7. Computational approaches to problem-solving (module 4)
8. Majority element finding
9. Palindrome check
10. Fibonacci sequence generation

****Medium Priority Topics****

1. Problem-solving strategies: trial and error, heuristics, means-ends analysis, backtracking
2. Python programming: variables, data types, modules, I/O operations, operators
3. Randomized simulation
4. Divide and conquer
5. Tabulation
6. Memoization
7. Recursion: definition, call stack, avoiding circularity
8. Matrix operations (lower triangular, upper triangular, or diagonal)
9. Random walk simulation
10. Population growth calculation

****Low Priority Topics****

1. Artificial intelligence
2. Machine learning
3. Data structures
4. Optimization techniques
5. Simulation methods
6. Problem-solving strategies
7. Computational complexity
8. Algorithmic thinking
9. Programming principles

10. Data analysis
11. Statistical modeling
12. Optimization algorithms
13. Computational geometry
14. Algorithmic complexity
15. Fibonacci numbers

****Day-wise Hourly Revision Schedule****

Day 1-5: High Priority Topics (5 hours/day)

- Day 1: Recursion (2 hours), Dynamic programming (2 hours), Greedy algorithms (1 hour)
- Day 2: Brute-force and divide-and-conquer approaches (2 hours), Algorithmic thinking and its practical applications (2 hours), Pseudocode and flowchart representation of algorithms (1 hour)
- Day 3: Computational approaches to problem-solving (2 hours), Majority element finding (2 hours), Palindrome check (1 hour)
- Day 4: Fibonacci sequence generation (2 hours), Randomized simulation (2 hours), Divide and conquer (1 hour)
- Day 5: Tabulation (2 hours), Memoization (2 hours), Recursion: definition, call stack, avoiding circularity (1 hour)

Day 6-10: Medium Priority Topics (5 hours/day)

- Day 6: Problem-solving strategies (2 hours), Python programming (2 hours), Randomized simulation (1 hour)
- Day 7: Divide and conquer (2 hours), Tabulation (2 hours), Memoization (1 hour)
- Day 8: Recursion: definition, call stack, avoiding circularity (2 hours), Matrix operations (2 hours), Random walk simulation (1 hour)
- Day 9: Population growth calculation (2 hours), Fibonacci numbers (2 hours), Greedy algorithms (1 hour)
- Day 10: Algorithmic thinking (2 hours), Programming principles (2 hours), Data analysis (1 hour)

Day 11-15: Review and Practice (5 hours/day)

- Day 11: Review high priority topics (2 hours), Practice problems (2 hours), Pseudocode and flowchart representation of algorithms (1 hour)
- Day 12: Review medium priority topics (2 hours), Practice problems (2 hours), Recursion: definition, call stack, avoiding circularity (1 hour)
- Day 13: Review low priority topics (2 hours), Practice problems (2 hours), Dynamic programming (1 hour)
- Day 14: Practice problems (5 hours)
- Day 15: Final revision and practice problems (5 hours)

****Final Revision Phase****

- Review all the high, medium, and low priority topics
- Practice as many problems as possible
- Focus on the most important and challenging topics
- Get enough sleep and stay hydrated to ensure mental clarity and focus during the exam.